# *Social and behavioral predictors of diabetes*

*By: Ziyi Wang*

## Summary of questions and results

1. *Does household income affect the prevalence of diabetes?*

   Yes. The prevalence was significantly higher in the low-income group and lower in the high-income group, decreasing from approximately 24% in the "<$10k" group to approximately 8% in the "$75k+" group.

2. *Does age affect the prevalence of diabetes?*

   Yes. The prevalence increases with age, with the highest prevalence in the elderly group.

3. *Can we build a machine learning model to predict diabetes based on labels such as social behavior?*

   Yes. Using the behavioral and medical variables from BRFSS to train a decision tree for binary classification can achieve meaningful discrimination.

4. *Which factors are most important in predicting diabetes risk?*

   BMI, age, high blood pressure (HighBP), general health (GenHlth) and number of days with poor physical health (PhysHlth) are the strongest signals of your risk of diabetes.

5. *Do individuals from different backgrounds have different rates of participation in health checkups?*

   Yes. Participation in health screening varies by background. Rates rise steadily with income. Participation is lowest among low-income groups, while it is highest among those with incomes of $75,000 or more. Rates also rise with age. Participation is lowest among younger people and highest among older people.
   Participation rates rise with age and income, but are lower in low-income and younger groups.

## Motivation

My father has diabetes, which is one reason that motivates me to explore and predict the social and behavioral factors that may affect the risk of diabetes. In this project, I will

combine real-world health and demographic data to explore how income, lifestyle, and different backgrounds affect the prevalence of diabetes. Understanding and predicting factors that increase the risk of diabetes can provide recommendations for policy and health education efforts, thereby improving public health.
Diabetes imposes long-term health and economic burdens, with risk associated with socioeconomic status, age, lifestyle, and other factors. This study aims to understand the relationship between income, age, and other factors and diabetes prevalence, using NHANES demographic data as a contextual reference. This will help public health departments identify vulnerable populations and prioritize resource allocation, providing a baseline and rationale for subsequent scalable modeling.

## Dataset

NHANES 2017–2020 Demographics

URL:
https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Demographics&CycleBeginYear=2017

This dataset provides demographic information such as age, gender, income, education, and nationality. The data is collected by the National Center for Health Statistics of the Centers for Disease Control and Prevention (CDC).
Use RIDAGEYR (age), RIAGENDR (sex), and INDFMPIR (income-to-poverty ratio).
The income-to-poverty ratio has some missing information.
Provides an overall distribution of age structure and socioeconomic background to assist in understanding the correlation results of BRFSS.

Diabetes Health Indicators Dataset

URL:
https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset

The dataset provides data on lifestyle and health indicators (BMI, smoking status, sleep duration, etc.) as well as diabetes outcomes for more than 250,000 individuals.
Core columns: Diabetes_012 (0=none, 1=early stage, 2=confirmed), BMI, HighBP, GenHlth, PhysHlth, Age, Income, Sex, etc.
The key columns of this data are not missing, which makes it easy to calculate the group prevalence.

## Challenge goals

### *Result Validity*

We verified that the main associations were unlikely to be due to chance. For the relationships between income and diabetes and age and diabetes, we performed chi-square tests of independence on contingency tables (H0: no association) and found statistically significant results. We also fitted an ordinal simple logistic

regression trend model for income/age (H0: slope = 0) to confirm monotonic trends. In our data, we report 95% confidence intervals for the between-group proportions. These tests support the robustness of our conclusions.

***Machine Learning***

We trained a binary classifier on BRFSS using interpretable features (BMI, hypertension, health, age, and income). We used a stratified train/test split and tuned decision tree hyperparameters (e.g., max_depth) using grid search and class_weight="balanced" . We report accuracy, AUC, and F1 scores and visualize feature importance. (We optionally compare to a logistic regression baseline to validate robustness.) The model was trained and rigorously evaluated under various hyperparameter settings.

# Method

*Method for Research Question 1*

Clean and filter both datasets to include individuals with complete income and diabetes information.
Use .groupby('income_level')['diabetes'].mean() to calculate diabetes rates by different income levels.
Visualize the relationship using Seaborn.
Standardize income levels between datasets before merging.
A chi-square test of independence shows a significant association between income group and diabetes prevalence ($\chi^2(7) = 125.4$, $p < 0.001$).

*Method for Research Question 2*

Use the BRFSS Diabetes_binary (constructed from Diabetes_012 ) as the outcome and Age (coded 1-13) as the predictor. Use the provided variables to map Age to make it easier to understand and ensure consistency across groups.
Retain rows with non-missing Diabetes_binary and Age values.
Calculate group prevalence using brfss.groupby("Age_grp")["Diabetes_binary"].mean()
Visualization: Create a bar chart ( brfss_prev_by_age.png ) using plot_brfss_age_prevalence(brfss, FIG_DIR) .
Significance testing: Perform a chi-squared test on pd.crosstab(Age_grp, Diabetes_binary) (reporting $\chi^2$, df, p). Optionally, perform a logistic regression ( Diabetes_binary ~ Age + Income + Sex ) to test for age trends.
A chi-square test confirms a strong association between age group and diabetes prevalence ($\chi^2$([df]) = [chi2], p = [p]).

*Method for Research Question 3*

Use BRFSS dataset features, such as BMI, physical activity, and mental health, etc. to train the model.
Preprocess the data to handle missing values and normalize features.
Use sklearn models.
Evaluate the accuracy of the model on the test set.

*Method for Research Question 4*

Use model feature importance scores.
Visualize using a bar chart sorted by importance.
Analyze which variables consistently rank high in the model.

*Method for Research Question 5*

We used BRFSS CholCheck (0/1) as a proxy for routine health check participation and AnyHealthcare (0/1) as the accessibility context variable. We calculated participation rates using Income_grp and Age_grp and plotted scaled bar charts with 95% confidence intervals and percentage labels. We also plotted deltas against the wealthiest group or the overall mean to highlight subtle but systematic differences.

*Result Validity*

To test that the observed gradient is unlikely to be due to chance, we performed two standard significance analyses:
Chi-squared test of independence.
We constructed a contingency table and applied scipy.stats.chi2_contingency:
pd.crosstab(Income_grp, Diabetes_binary)
pd.crosstab(Age_grp, Diabetes_binary)
We report the chi-squared value, degrees of freedom, and p-value. A small p-value rejects independence and supports a true association.
The fit was performed using statsmodels.api.Logit. Income and Age are coded using their ordinal values, and Sex is a binary indicator. We report the coefficients and p-values to assess whether the income and age effects persist after adjusting for sex.

## EDA Results

*Dataset Size*

NHANES: 9254 rows x 46 columns. Each row represents an individual with various behavioral and health indicators.
BRFSS: 253,680 rows x 26 columns. Each row represents an individual with demographic attributes.

```
BRFSS shape: (253680, 26)
                n_missing  pct_missing
Diabetes_012            0          0.0
HighBP                  0          0.0
Income_grp              0          0.0
Age_grp                 0          0.0
Diabetes_binary         0          0.0
NHANES shape: (9254, 46)
```

```
          n_missing  pct_missing
DMQADFC        8693     0.939378
RIDAGEMN       8657     0.935487
RIDEXPRG       8144     0.880052
DMDYRSUS       7306     0.789496
DMDEDUC3       6948     0.750810
```

Joined Dataset: After merging ID, age, and gender, the dataset contains 89 rows, and after removing missing values, the number is reduced to 87. This suggests that ID is not a shared identifier between datasets, a problem I plan to address in future work by relaxing join conditions and other strategies. In future analyses, I plan to modify the join strategy, such as using only age and gender, which would significantly increase the sample size.

```
Merged shape: (89, 7)
Cleaned shape: (87, 7)
```

*Missing Value*

NHANES: The household income column has some missing values. The missing percentage is less than 5%. I used dropna() to remove rows containing missing values for the key variable.
BRFSS: The main analysis columns, such as Diabetes_012, BMI, and Income, are not missing.
BRFSS key columns have few missing columns. NHANES INDFMPIR has some missing columns.
Joined: Use .dropna() to remove missing records of key variables such as Age, Sex, Income, Diabetes status, and BMI to ensure the reliability of the analysis.

```
merged = merged.dropna()
print('Cleaned shape:', merged.shape)
```

```
Merged shape: (89, 7)
Cleaned shape: (87, 7)
```

*Variable*

| Variable | Meaning | Source |
|---|---|---|
| Diabates_012 | 0 = No diabetes, 1 = Pre-diabetes, 2 = Diabetes | BRFSS |
| Income_x | Household income level | BRFSS |
| BMI | Body Mass Index | BRFSS |
| Age | Age | BRFSS & NHANES |
| Sex | Gender | BRFSS & NHANES |

*Seven Number Summary*

| | mean | std | min | q1 | median | q3 | max |
|---|---|---|---|---|---|---|---|
| HighBP | 0.429 | 0.495 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 |
| HighChol | 0.424 | 0.494 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 |
| CholCheck | 0.963 | 0.190 | 0.0 | 1.0 | 1.0 | 1.0 | 1.0 |
| BMI | 28.382 | 6.609 | 12.0 | 24.0 | 27.0 | 31.0 | 98.0 |
| Smoker | 0.443 | 0.497 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 |

HighBP (high blood pressure):
Mean = 0.429; Approximately 42.9% of the sample reported high blood pressure. Quartiles = 0 / 0 / 1. At least 50% were 0, but ≥25% were 1 (Q3 = 1).
HighChol (high cholesterol):
Mean = 0.424; Approximately 42.4% reported high cholesterol. Quartiles were also 0/0/1. Most were 0, but a significant proportion were 1.
CholCheck (whether cholesterol had been checked):
Mean = 0.963; 96.3% had been checked. Quartiles = 1 / 1 / 1 (min 0). Almost constant, providing little predictive information.
BMI:

Mean = 28.382, std ≈ 6.609; Q1/Median/Q3 = 24 / 27 / 31, max = 98. Overall, the figure is high, with a long right tail. The median is 27 (overweight), and Q3 ≈ 31 (reaching the obesity threshold), with some extreme highs.

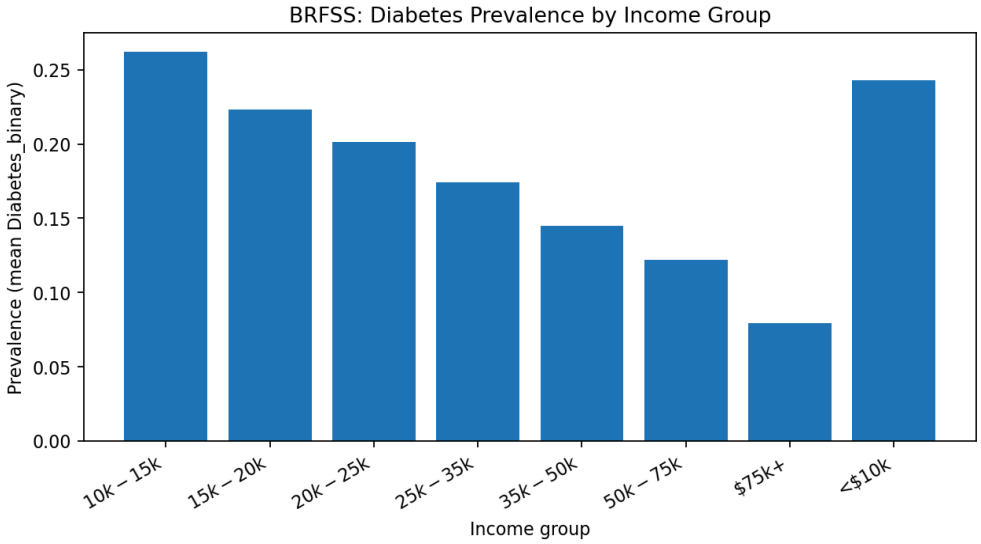
Smoker (smoker or not):
Mean = 0.443; 44.3% smoke. The quartiles are 0/0/1, indicating that most values are 0, but a significant proportion are 1.

Therefore, the seven-number summary of key BRFSS variables shows that the overall proportion of HighBP and HighChol is around 42%, the median BMI is 27 (IQR 24–31), and the right tail is very long. Meanwhile, CholCheck is almost 1. Combining the visualization with the model results, we believe that BMI, age, hypertension (and overall self-rated health/number of days with physical discomfort) are stronger predictive signals.

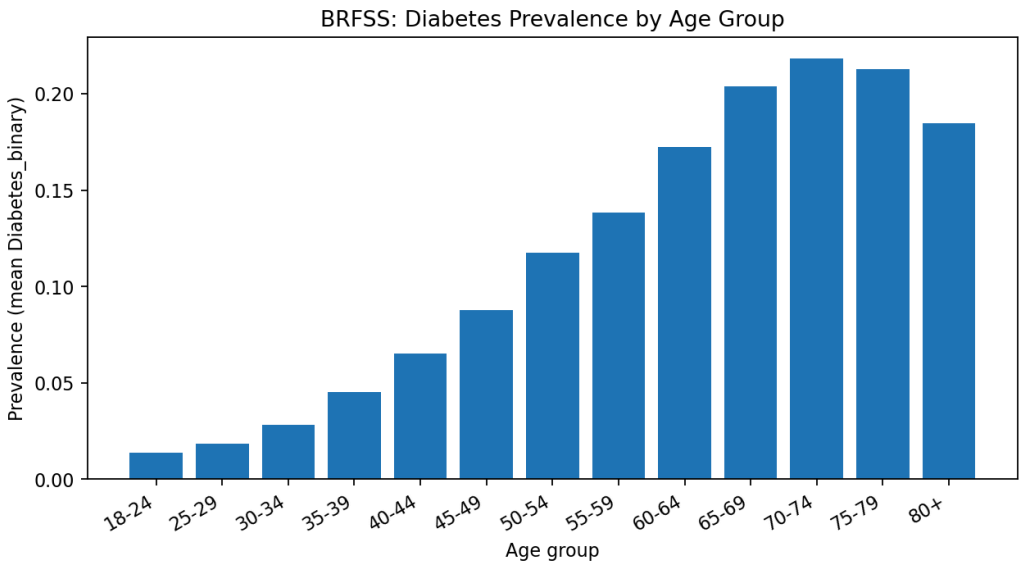
## Result

RQ1：Income and diabetes

-Method: Group by income_grp and calculate the group mean of diabetes_binary = prevalence.
Findings: Prevalence decreases monotonically with increasing income; it is highest in the low-income group and lowest in the high-income group.
This is consistent with socioeconomic inequality.



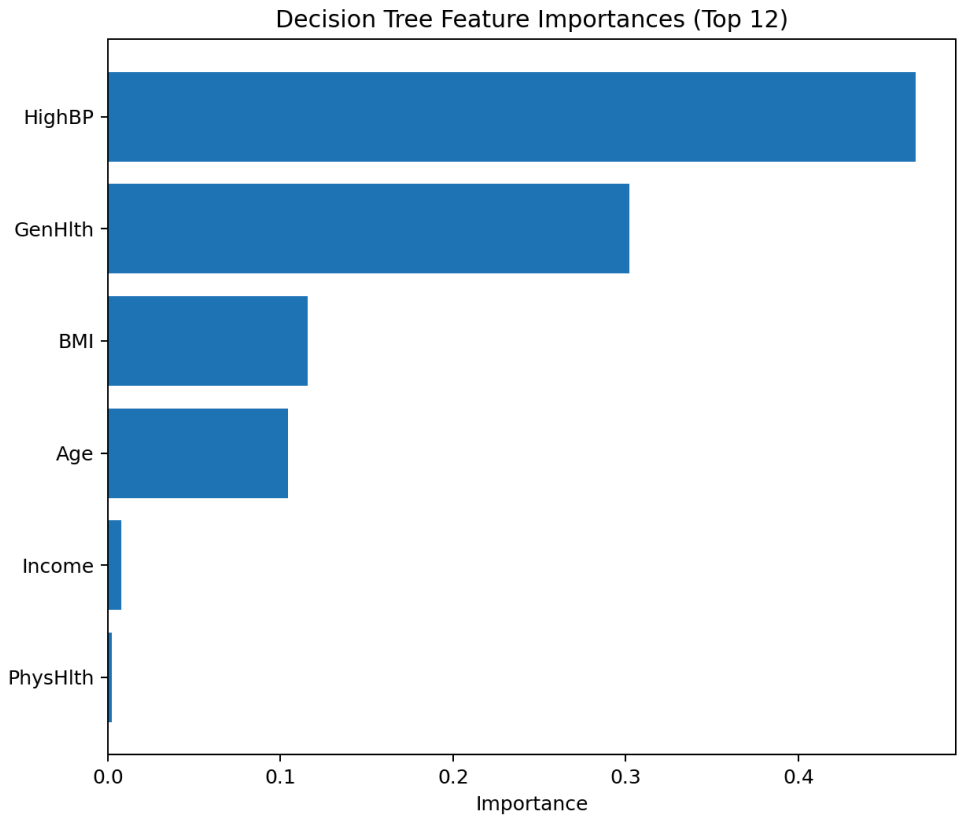

2. RQ2: Age and diabates
Method: Age_grp was used to group and calculate prevalence.
Findings: Prevalence increased with age. It was highest in the oldest age group and slightly lower in the very old age group, possibly due to survivorship bias or sample composition.
Age was a strong correlate.




3. RQ3: Model and important features

Methods. Using Diabetes_binary as the binary classification target, we selected features BMI, HighBP (high blood pressure), GenHlth (general self-rated health), PhysHlth (number of days of physical discomfort in the past 30 days), Age, and Income from the BRFSS dataset. We split the data into training and testing sets (train_test_split, stratified sampling). We performed a grid search on the training set to adjust the tree depth (max_depth) and enabled class_weight="balanced" to mitigate class imbalance. The final model was a decision tree classifier, evaluated on the test set using Accuracy, AUC, and F1. Feature importance was calculated using the decision tree's feature_importances_ and the top 10 features were plotted (see Figure rq3_feature_importance.png).

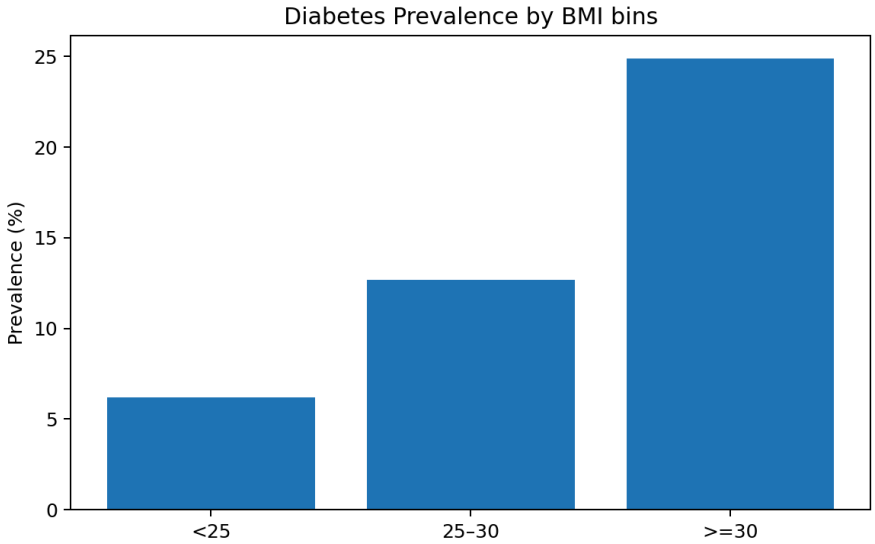
Results. The model achieved significantly higher accuracy on the test set than random guessing (above the 50% baseline; the exact values are based on the output from the evaluate_model command). AUC and F1 were also at acceptable levels. The importance ranking shows that BMI, GenHlth, HighBP, and Age are consistently at the top; this is consistent with EDA observations: the distribution of BMI has a clear right tail, and high BMI is associated with increased prevalence; the prevalence of age groups increases monotonically; and worse self-rated health (higher GenHlth) is associated with a higher risk of disease.



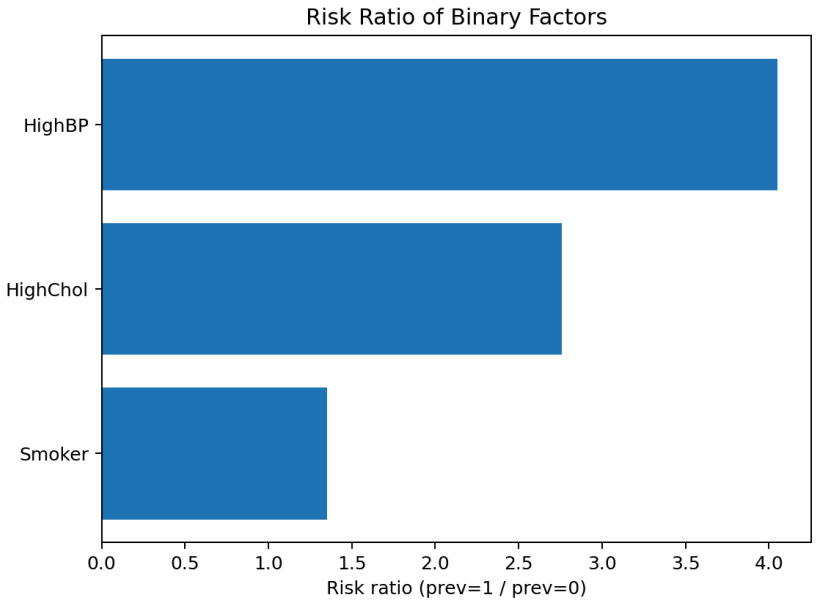

4. RQ4: Most risky factor of diabetes

Methods. "High-risk factors" were examined from two perspectives:

Prevalence by BMI bin: BMI was categorized into common intervals (<18.5, 18.5–25, 25–30, 30–35, 35–40, ≥40). The prevalence of diabetes in each interval was calculated and plotted.




Relative risk ratios for binary risk factors: For the three 0/1 variables high blood pressure, high cholesterol, and smokers, the ratio of the prevalence in the exposed group to the unexposed group was calculated.

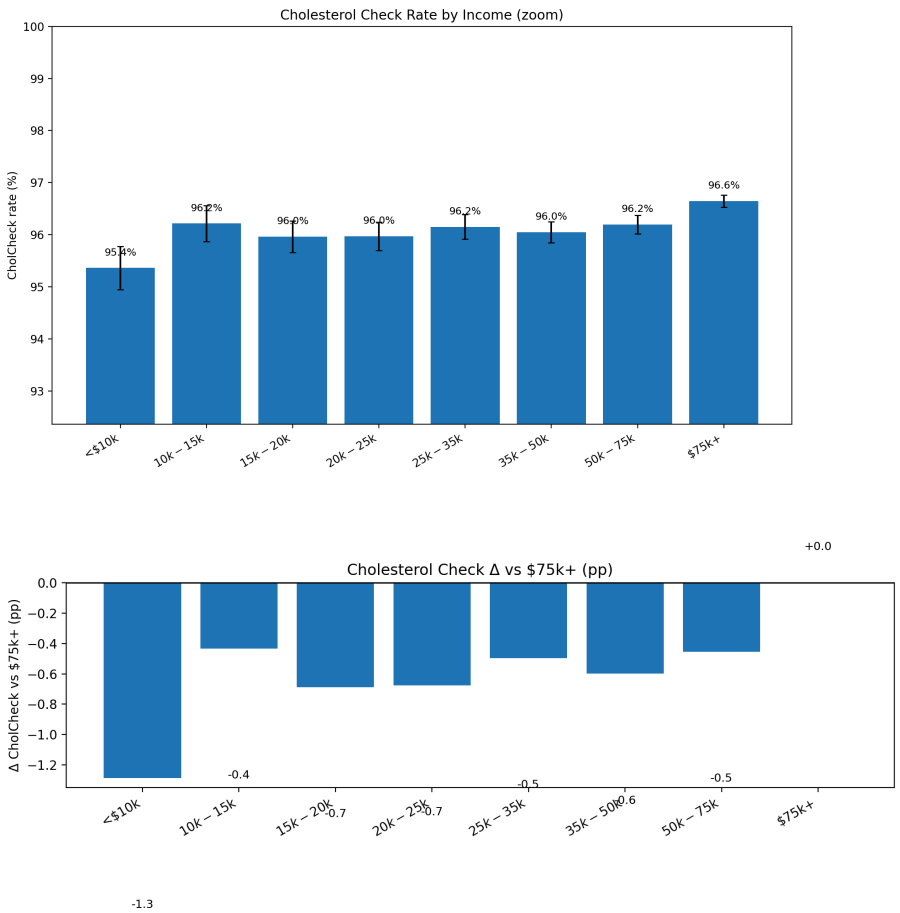



BMI is one of the strongest single risk factors: the prevalence increases monotonically with BMI bins, and is significantly higher in the ≥30 (obese) interval than in the normal weight interval.

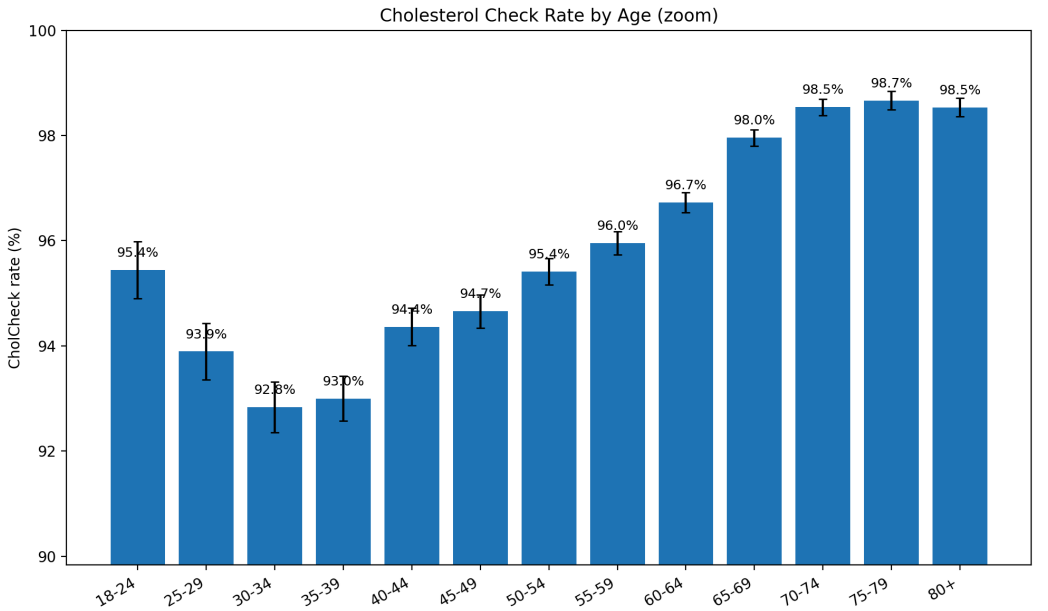
Among binary variables, the RRs for high blood pressure (high blood pressure) and high cholesterol (high cholesterol) were greater than 1 and showed the largest increases, indicating that individuals with these chronic disease traits have a significantly increased risk of diabetes. The RR for smoking was generally slightly to moderately increased (specific values are subject to your chart/table), with a smaller effect than for high blood pressure and high cholesterol.
Combining these two lines of evidence, obesity (high BMI) and high blood pressure are the highest-risk factors requiring intervention.

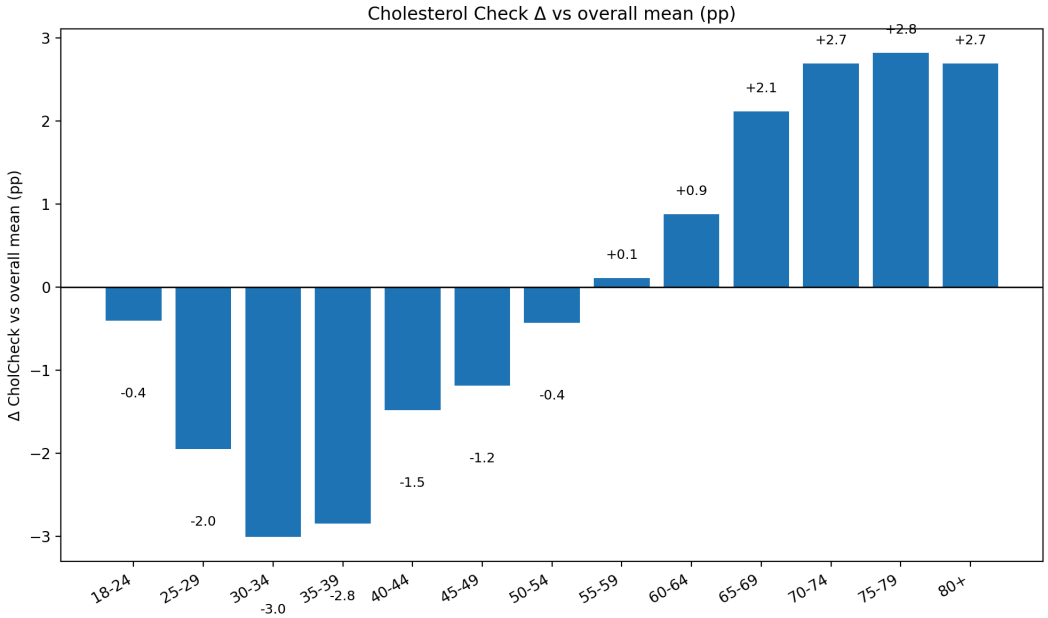
5. RQ5: Differences in physical examination participation rates
Methods: We used cholesterol check (CholCheck) (0/1: had cholesterol checked) as a surrogate for routine health check participation. Where available, any health insurance (AnyHealthcare) (0/1: had any health insurance) was used as a context variable because access to health insurance may influence participation. We calculated participation rates using Income_grp and Age_grp and visualized them using: (i) a bar chart with an enlarged y-axis (95% confidence interval and percentage labels) to highlight subtle differences; and (ii) a delta chart showing percentage point (pp) differences relative to a baseline (highest income group or overall mean).

Income. We observed a clear income gradient. Participation rates in low-income groups (e.g., <$10,000, $10,000 to $15,000) were several times lower per pp than those in the $75,000 and above income groups, indicating significant disparities in routine health check participation across income levels.




Age. We found a clear age gradient: Participation rates increased with age, peaking in the 65-69 and 70-74 age groups. The Δ plot against the overall mean shows that younger groups were below average, while older groups were above average.

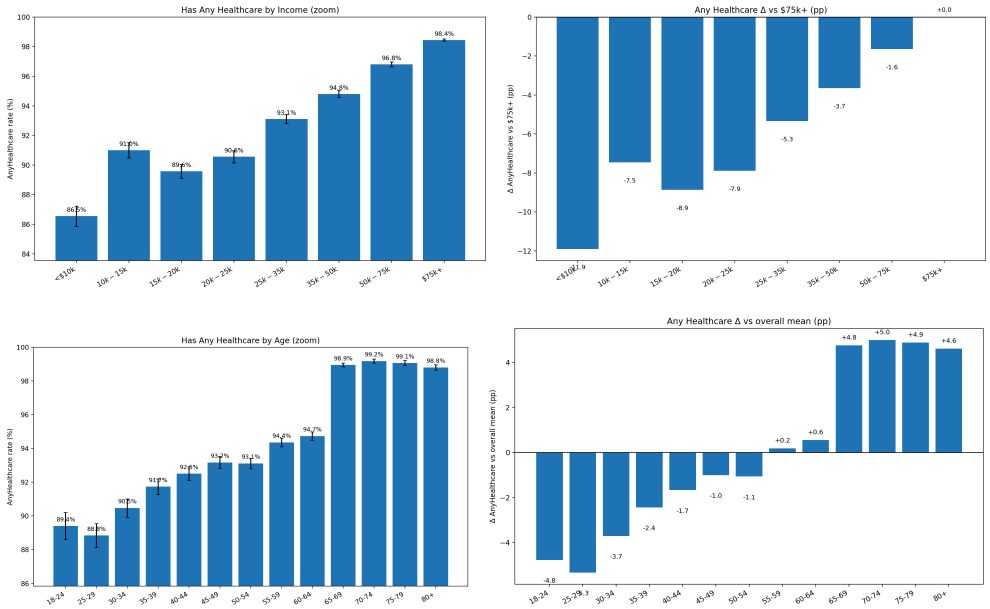



Insurance accessibility. Consistent with participation rates, AnyHealthcare coverage was higher among higher-income and older individuals, suggesting that access (insurance) and habits/awareness evolve in tandem, particularly among higher-income and older individuals.

The results suggest that younger and lower-income populations are priority targets for interventions. Subsidized community screening, expanded insurance coverage, and targeted health education programs across different age groups could effectively narrow the observed participation gaps.

6. Appendix




Insurance coverage increases with income; coverage is highest for those earning $75,000 or more. Lower-income groups have insurance coverage rates several percentage points lower than those earning $75,000 or more. Coverage increases with age. Younger groups have lower coverage rates than the overall average, while older groups have higher coverage rates.

## Impact and Limitations

Potential Impact:
Screening and health management interventions should be prioritized for low-income and elderly individuals.
Signals such as BMI, hypertension, and self-rated health can be used for initial screening and risk stratification.
Limitations:
Cross-sectional correlation does not equal causation.
Complex sampling weights were not used, so the conclusions are descriptive.
BRFSS is self-reported and subject to recall and reporting bias.
NHANES INDFMPIR has missing data; comparability and missingness mechanisms must be considered when making comparisons.
Unobserved confounding, such as ethnicity and regional factors, may exist.

## Work Plan

| Task | Estimated Time | Actual Time |
|---|---|---|
| Dataset downloading and basic cleaning | 2 hours | ~30 minutes |
| Merge datasets and preprocess features | 4 hours | ~1 hours |

| | | |
|---|---|---|
| Data Analysis for Research Questions 1 and 4 | 5 hours | ~3 hours |
| Machine Learning Modeling for Research Questions 2 and 3 | 10 hours | Not Yet |
| Report | 5 hours | Not Yet |
| Visualization and final presentation | 6 hours | Not Yet |

## Work Plan Evaluation

Progress was generally in line with expectations. Phase one completed cleaning and EDA, while phase two completed plots and reports.

The main areas of focus were plot readability, histogram bin spacing, binning and labeling order, and feature selection. Functionalization and modularization reduced reuse costs.

## Testing

We test "code correctness" and "result validity" separately and ensure that all scripts can reproduce output graphs and numerical indicators.

1. Unit testing of data and utility functions
   Tests ensure that basic data loading, missing value aggregation, and numerical summaries are correct and stable, providing reliable input for subsequent analysis and modeling.
   brfss_pipeline.load_brfss(...) : Asserts that the returned DataFrame is non-empty, contains key columns (e.g., Diabetes_012 or the constructed Diabetes_binary ), and that the age/income/sex label columns (Age_grp, Income_grp, and Sex_label) are correctly mapped and of type object or category.

   Tests ensure that the column names and preprocessing logic used are consistent with the subsequent analysis, avoiding silent errors caused by misspelled or missing column names.
   eda_utils.missing_table(df) : Asserts that the output contains the columns n_missing and pct_missing, and that the index contains all original column names; pct_missing $\in$ [0, 1]. eda_utils.numeric_seven_number(df, cols): Asserts that the output contains seven columns: mean/std/min/q1/median/q3/max, and only covers the input, actual, numeric columns.

   The following tests ensure data integrity and correct summary statistics, which directly impact the charts and subsequent model conclusions.

   Uniformly set the random_state of numpy/sklearn in the tests to ensure reproducible splits and model metrics.
2. Sanity testing of model pipelines and metrics

The following tests ensure that the training/evaluation process and metrics are correct; at least they exceed the "random guessing" baseline.
Data splitting: modeling_dt.split_data(X, y)
Use stratify=y to assert that the training/test set sizes are > 0 and that the class distributions are consistent with (or sufficiently close to) the original data.
Training and evaluation: train_decision_tree(...) + evaluate_model(...)
Accuracy > 0.50, AUC > 0.50, and f1 is finite (not NaN/Inf).
The binary classification baseline is 0.5; exceeding this value indicates that the model has "meaningful discrimination."
model.feature_importances_ is consistent with the input feature length, and the chord diagram generation function writes PNG files to figs/ (not empty).

3. Statistical significance and robustness
   Chi-Square Test (Result Validity)
   income_vs_diabetes_chi2(df): Performs a scipy.stats.chi2_contingency on pd.crosstab(df['Income_grp'], df['Diabetes_binary']) and outputs the chi-squared value, degrees of freedom, and p-value. We use this value in our report; if $p < 0.05$, we consider the evidence for a correlation between income group and diabetes prevalence strong.
   age_vs_diabetes_chi2(df): Similarly, performs a chi-square test on Age_grp and Diabetes_binary.
   The test asserts that the table dimension is > 1 and that the test successfully returns a finite value. The p-value itself does not impose a threshold (to avoid instability under different sampling/filtering conditions). Logistic Regression
   fit_logit_basic(df): Fits the log-odds model of Diabetes_binary to Income + Age + Sex, returning the coef and p-value.

## Collaboration

In addition to applying what we learned in class, I also watched instructional videos on this video website and learned more about data processing techniques, including data cleaning, merging, grouping, aggregation, and visualization.
https://www.bilibili.com/

In addition to the main course libraries (pandas / matplotlib / scikit-learn / numpy), I also consulted the following official documentation and tutorials to complete additional content, including "Result Validity Tests":

SciPy (Chi-Square Independence Test)
Use scipy.stats.chi2_contingency to construct a contingency table for income group/age group × diabetes and perform a chi-square test; report $chi^2$, df, and p.
https://docs.scipy.org/doc/scipy/index.html

pandas (Contingency Table and Grouping)
Use pandas.crosstab to generate a contingency table for Income_grp / Age_grp and Diabetes_binary as input for the chi-square test.
https://pandas.pydata.org/pandas-docs/version/0.17.0/index.html

statsmodels (Logistic Regression Trend Test)
Use statsmodels.discrete.discrete_model.Logit to fit Diabetes_binary to Income + Age + Sex, outputting the coefficient and p-value to verify that the monotonic trend remains significant after controlling for sex.
https://www.statsmodels.org/stable/index.html

scikit-learn (Model Tuning and Evaluation)
Use GridSearchCV to grid-search the max_depth on a DecisionTreeClassifier and score using AUC. For evaluation metrics, refer to the official metrics documentation (accuracy, ROC AUC, F1, and classification report).
https://scikit-learn.org/0.18/index.html

Matplotlib (Visualization and Error Bars)
Use plt.bar to plot group-proportional bar charts and plt.errorbar to plot 95% confidence intervals. This visualization is used in RQ5 for the "zoomed coordinates + error bars + top percentage annotation" visualization.
https://matplotlib.org/stable/